

XGBoost Regression model

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SETTING A WORKING DIRECTORY

```
setwd("C:/Users/ADMIN/OneDrive/Desktop/caleb makini")
```

loading libraries

```
library(xgboost)
library(rio)
library(caret)
library(ggplot2)
library(dplyr)
library(Matrix)
library(Metrics)
```

import dataset

```
set.seed(123)
n <- 500
HouseSize <- round(rnorm(n, 1800, 400))
Bedrooms <- sample(2:6, n, replace = TRUE)
Bathrooms <- sample(1:4, n, replace = TRUE)
Age <- sample(1:50, n, replace = TRUE)
Garage <- sample(0:3, n, replace = TRUE)
DistancedCBD <- round(runif(n, 2, 30), 1)
CrimeRate <- round(runif(n, 10, 80), 1)
SchoolRating <- sample(1:10, n, replace = TRUE)
LotSize <- round(rnorm(n, 600, 120))
HousePrice <- 5000 + HouseSize * 180 + Bedrooms * 12000 + Bathrooms *
15000 - Age *
1500 + Garage * 8000
-DistancedCBD * 2500 - CrimeRate * 900 + SchoolRating * 10000
```

[1]	27790	8670	10410	-59590	-41320	3520	-97960	-61310
-36640								
[10]	-60220	21970	-18220	16300	-45620	-58980	41620	-4600
-27730								
[19]	-84310	-68670	-42300	-1470	11710	-30700	-56720	-73670
-45800								
[28]	-77010	-76090	-1050	-2870	-5670	-86490	-92030	23200
18950								
[37]	-39890	51700	-19300	-45440	-67590	-38180	-31870	-81000
-67340								
[46]	44610	19830	-35220	-78190	-21350	16210	-22910	-69230
-5540								
[55]	-7320	-93430	-46420	-50010	27000	22710	-60630	-23510
-43740								
[64]	7740	17910	37010	13310	-92580	-24300	-35910	-69960
6070								
[73]	-46780	-73910	-57820	-55050	-66470	-1700	8690	-80720
-9490								
[82]	-60480	-68870	23790	-26090	-35160	15260	-85210	-51880
-52980								
[91]	-83990	-16410	-47010	-470	45780	-88010	8540	-71470
-48430								
[100]	-50510	-37790	-29800	-60060	44940	-2940	-74040	-24640
-37690								
[109]	-47010	22260	-82200	-86080	-36290	470	-45930	-14910
-49150								
[118]	-68490	-53170	-71940	-119210	-109960	1980	-24480	-4260
-46380								
[127]	-47920	39160	-800	-36100	-3060	250	51650	-97330
1320								
[136]	6730	-62150	-26080	-52470	-24560	-4210	-84330	-56010
-51500								
[145]	65420	480	42450	14510	-14950	-10410	-67810	-54300
-65790								
[154]	590	-71230	-75710	-48970	-50270	-49200	-58210	-37970
-18410								
[163]	-77240	-77580	33150	-90980	-71630	-53380	45080	29870
-86570								
[172]	-59570	-69640	22300	-57750	19480	-1090	2670	-16360
-40520								
[181]	-15180	-36840	-46010	32280	1970	-84540	-26950	15410
-53460								
[190]	-25200	33180	-115370	-34310	-3970	-74850	-31770	16260
54600								
[199]	-20690	-99080	18260	-54780	-43850	-78660	-48340	-19010
-37660								
[208]	-35560	-28820	1800	16040	-10890	47970	3970	-51780
-109190								
[217]	-10380	2210	-11850	-11180	-26170	-32650	9780	13430

63840								
[226]	-4620	-15480	-105120	-50410	-59910	5140	-1010	-21910
-73640								
[235]	-4980	10520	26590	-28860	20970	-58440	44260	15720
-66760								
[244]	-114260	14580	-38430	19290	42940	-40040	-16840	-102750
-87720								
[253]	15260	-49740	-30820	-22240	-46670	-34770	-34290	6640
-34990								
[262]	-53080	8890	-54790	23180	-32250	-82690	-18700	-43450
-23910								
[271]	29430	-27270	-11930	-30740	-33010	-23480	-47340	-87080
5450								
[280]	-11380	6060	-87290	-87520	910	-92560	-28910	-5830
-39030								
[289]	5730	-20920	-84970	-34730	-13990	1910	-17120	-77750
-120270								
[298]	9420	-72760	2630	34000	-102570	5520	19120	-32350
17550								
[307]	-630	-46930	50600	-24760	-34920	-87230	-27930	-62750
7140								
[316]	6690	-68520	-43720	-99490	-33240	72660	-64940	-15200
-740								
[325]	-11280	-7930	16710	-21260	-10560	-3220	3850	-1020
33710								
[334]	-39060	-43800	3800	36050	-36340	37070	-69470	-46190
-59970								
[343]	19660	-42380	-37090	19400	-23310	-22250	-66510	-85030
-11110								
[352]	-69350	-90800	9950	-62480	-25870	-5660	-11540	-23470
-5080								
[361]	15820	-42080	21170	-940	-56470	14840	-36990	-56170
-9050								
[370]	8500	-22590	-74220	-60910	-42690	-39900	-37440	-88930
-53340								
[379]	-69290	5170	-33590	-16980	-56930	49710	3820	-24910
-25080								
[388]	6320	-36120	15630	6250	-10110	-23820	-23200	22830
-66360								
[397]	-36780	12940	-21700	-69300	-270	-11890	-71360	17010
-550								
[406]	-74450	-56690	-7920	-64650	-38480	-30600	-89280	24730
-74680								
[415]	-17250	-26240	-2970	1530	54240	-38890	-29100	30
58240								
[424]	35490	43560	-78660	-9110	51300	-28030	-32430	-9360
15070								
[433]	-65380	-47130	-77220	-9080	-45660	-77760	1450	-34390
-31750								
[442]	-1040	6610	12270	-41630	-58500	-24780	-66230	-62000

```
43700
[451] -3150 -54930 -63370 -106010 -114700 -92790 4140 -21940
-86570
[460] 71790 12750 -74110 52130 15160 -58420 -27700 16550
3970
[469] -63540 20280 -17230 -81010 -128750 4200 -42570 -36880
-14570
[478] -46760 -48540 56100 51640 -57500 42420 22130 -52210
-76410
[487] -12070 13650 -1070 -18180 31240 -25480 -70720 -64760
5030
[496] -38710 -99080 -81230 -99320 -23870
```

```
+LotSize * 60 + rnorm(n, 0, 25000)
```

[1]	30384.1526	-12682.4402	40635.9592	45856.1540	-3697.4673	58
	92.4845					
[7]	86699.9278	40967.6810	30309.8232	31457.3136	51856.1940	405
	31.0265					
[13]	14612.2012	34248.0983	38127.8823	30708.2651	18347.5282	633
	92.5381					
[19]	40205.5539	19373.3293	51741.2184	49435.7120	30160.5105	541
	82.0467					
[25]	13040.5251	18368.9937	10301.1324	9644.7391	36595.0337	406
	20.6200					
[31]	-805.1383	31683.5468	-18401.1530	17811.5687	57886.6819	14
	54.6026					
[37]	14336.6995	24503.4635	46186.7683	68222.7291	108763.5182	512
	22.3310					
[43]	46846.3382	82978.1029	20016.8924	37266.2649	26611.7537	-32
	82.2011					
[49]	-20265.9164	67069.7344	21805.1064	35652.9786	22936.1059	355
	21.8149					
[55]	45738.5131	26384.7191	41467.6173	41364.8229	37530.7981	190
	26.4132					
[61]	29144.7480	481.8413	37047.2021	1671.8554	70728.1361	-325
	83.0976					
[67]	46392.3534	27099.3194	82078.7202	22758.4142	43665.9340	408
	72.3492					
[73]	31451.8102	43312.9959	84555.3433	57627.7206	43310.6037	900
	56.6555					
[79]	52979.9437	62878.3400	62445.7537	49747.6657	58397.2673	205
	57.9945					
[85]	42000.9016	28051.0657	26846.6741	82966.3305	20347.5600	221
	85.5147					
[91]	62990.4348	47724.4443	72343.1607	25676.0720	36218.9585	350
	06.5780					
[97]	16512.1590	39937.2012	25738.3215	55936.1831	27165.4104	305
	70.4699					
[103]	12663.8827	39699.9751	4521.5911	-54486.7946	77962.6586	539
	71.5424					
[109]	42628.2159	-8272.7066	51248.2279	40955.3568	27638.6240	629
	25.9691					
[115]	77454.2258	32045.9100	85413.4269	53869.0675	76089.6345	333
	42.0596					
[121]	15921.9286	8315.3644	44504.2271	34620.2859	36302.0888	108
	22.8143					
[127]	32780.7626	88027.7946	18591.0161	28847.7442	12374.1862	22
	21.7733					
[133]	13350.8700	26397.8175	25257.4609	13773.8923	41164.3269	722
	10.5136					
[139]	30894.9704	4399.9034	5313.7049	-15368.6912	-3402.9081	242
	74.4954					
[145]	9429.6135	2054.3472	103551.0913	83498.2771	93630.2325	621

32.3975						
[151]	32388.7864	15858.4826	8840.8434	23158.1513	68673.0232	556
61.7641						
[157]	53857.5049	26581.9643	-4160.9086	45420.0668	28354.7007	306
81.0423						
[163]	82684.3307	47422.2821	43909.8449	-8706.1605	25953.9632	587
19.7230						
[169]	50960.5248	41302.6842	63507.3410	29854.7503	23754.9494	339
70.8467						
[175]	55382.8232	38626.2967	46907.5286	4335.4095	2723.3745	314
44.5902						
[181]	62209.5789	25028.6771	11604.4664	-39244.4042	37378.0219	209
28.6255						
[187]	64364.3727	66820.2931	22942.0239	55458.1345	51686.3557	284
75.1159						
[193]	3385.0154	37805.5899	67004.0386	78937.7964	1035.4836	625
42.1252						
[199]	25159.3957	66288.0157	37478.4034	20679.1684	47652.4874	566
35.0382						
[205]	38836.9259	50181.2346	50741.5103	46019.3500	37320.7540	542
70.2556						
[211]	44447.4823	58308.3210	70334.2125	49133.4333	37575.0418	247
83.2394						
[217]	37719.1887	10711.1271	32497.2853	17827.6357	36062.8546	99
82.5020						
[223]	22422.4471	38709.6789	50040.1488	25298.2580	58080.5583	-55
68.2360						
[229]	48608.9652	87005.7642	17823.8060	63389.6230	16462.7999	367
09.6241						
[235]	41570.0866	8409.6380	27409.4801	52965.5837	49332.7422	528
11.4795						
[241]	54058.7049	47506.3185	7379.1848	44344.5342	32022.8145	336
93.5625						
[247]	57854.0589	41703.8861	40299.7090	44037.1314	40705.4626	50
34.3703						
[253]	26001.2994	66754.4009	44743.0711	13395.1493	8346.0835	202
36.7437						
[259]	47143.9214	57268.9741	15545.4424	15262.8915	46415.7017	362
43.1598						
[265]	36759.7680	43853.3437	29341.8092	29024.3399	34316.4681	17
74.9891						
[271]	37712.8154	77101.0031	46126.5659	52658.2307	22601.6244	179
12.2806						
[277]	27797.1961	3353.9844	37682.3236	38740.9315	60405.1614	772
91.3510						
[283]	-7596.2359	50121.2865	59206.8518	36989.3548	47281.4833	-56
63.3119						
[289]	43819.8668	-56027.2048	36961.4576	81619.0864	62309.6757	-391
31.7345						
[295]	47635.6711	25096.9751	22774.7985	-15743.3364	38246.0125	216

91.9828						
[301]	43907.2372	36259.5450	9884.0611	43369.9084	62816.7268	400
57.9822						
[307]	54677.5397	27819.7343	72260.0204	36650.6930	21088.7028	637
79.8196						
[313]	78032.0545	49176.3909	50812.6665	71492.1496	23033.9777	596
44.2498						
[319]	72979.8459	44805.2394	57603.0501	55202.5548	67819.5502	162
69.7393						
[325]	73212.6548	-17043.6767	80842.5886	12536.5060	34995.2259	407
32.1294						
[331]	3782.1985	42267.6289	33397.1860	42080.8128	76517.3515	8
10.9837						
[337]	35705.2896	62651.1822	13835.8799	42186.2220	96822.1224	-227
43.6399						
[343]	40570.0534	24579.7859	47958.0346	13866.7869	42895.1401	191
94.8404						
[349]	37526.8610	58125.3720	40805.0938	-1897.6363	6762.8133	502
99.8774						
[355]	83523.7080	2489.3464	37904.4741	30780.2530	14565.8378	100
14.1794						
[361]	87969.9150	71878.3615	-1509.6679	-21539.3033	24025.8443	654
80.6951						
[367]	32540.3333	70831.6149	28516.7877	11241.3046	73637.0573	399
89.3314						
[373]	25300.6603	17121.0429	43880.1076	-23706.5472	63642.4283	549
30.7975						
[379]	62171.1298	28241.8595	35962.7107	5347.1832	41694.5089	276
93.6161						
[385]	70002.3231	-1785.5892	-18897.1346	28556.4796	-20423.9033	132
23.8345						
[391]	60780.8073	77348.2937	19893.0477	55971.2824	-9594.0965	843
02.5329						
[397]	40012.8205	5042.9939	41438.5980	23328.7766	36058.0758	386
04.3010						
[403]	68228.0887	26335.2227	35316.4928	44679.8385	48534.3619	808
09.9248						
[409]	77714.6614	59226.3838	-27236.9244	54223.1188	62155.5093	196
95.0718						
[415]	11256.7447	3331.2414	34546.3256	43125.7504	7341.2385	-144
53.4408						
[421]	36702.1269	9307.9015	13159.5919	60952.9172	21092.2200	552
56.6079						
[427]	30394.2756	18623.5572	74326.9255	64205.3839	68904.1323	202
44.9724						
[433]	7372.0103	42419.0737	56443.3063	102389.2156	86727.3232	200
57.9309						
[439]	6630.5677	53684.7241	88051.0623	16321.6509	19967.4598	807
12.3747						
[445]	87989.6746	36963.9104	3354.2429	15055.7368	67645.4710	775

```

43.3394
[451] 37403.0865 8933.8717 41952.1829 26961.9651 8646.6173 452
17.1043
[457] 74781.1077 32128.0107 46123.7550 51122.1258 59539.1523 385
19.4227
[463] 55206.9014 50975.0783 14608.9637 51221.4283 65212.4381 325
77.9284
[469] -32034.1103 77804.1097 20294.2363 30211.4023 64045.7685 -127
10.9972
[475] 10376.3910 21523.7323 897.9657 58542.2081 57772.4033 433
01.9033
[481] 24114.8544 20848.0006 26058.1732 25522.3408 36490.6398 610
11.5841
[487] 28485.5578 42114.7600 -23824.8188 34775.9243 11938.9536 562
03.3295
[493] 68705.6102 -27732.3845 54797.9380 15256.7902 25601.4745 558
22.5207
[499] 49749.1830 10484.4189

```

```

housing_data <- data.frame(HouseSize, Bedrooms, Bathrooms, Age, Garage,
  DistancedCBD,
  CrimeRate, SchoolRating, LotSize, HousePrice)
head(housing_data)

```

```

  HouseSize Bedrooms Bathrooms Age Garage DistancedCBD CrimeRate SchoolRating
1      1576         6         3   5      3          5.9       19.4
6
2      1708         3         1  23      0          12.4       33.7
7
3      2423         4         1   3      3          5.9       27.6
5
4      1828         2         2   6      0          17.9       27.6
1
5      1852         6         4  25      3          17.2       64.8
6
6      2486         4         3  19      2           6.6       22.2
4
  LotSize HousePrice
1      612     422180
2      449     328940
3      631     523640
4      501     379040
5      473     456860
6      687     532980

```

explore data set

```
str(housing_data)
```

```
'data.frame':  500 obs. of  10 variables:
 $ HouseSize   : num  1576 1708 2423 1828 1852 ...
 $ Bedrooms    : int   6  3  4  2  6  4  3  2  5  5 ...
 $ Bathrooms   : int   3  1  1  2  4  3  1  1  2  1 ...
 $ Age         : int   5 23  3  6 25 19 11 43 12 33 ...
 $ Garage      : int   3  0  3  0  3  2  1  2  0  2 ...
 $ DistancedCBD: num   5.9 12.4 5.9 17.9 17.2 6.6 28.5 19.5 3.8 7.1 ...
 $ CrimeRate   : num  19.4 33.7 27.6 27.6 64.8 22.2 51.9 58.4 74.6 58.
 3 ...
 $ SchoolRating: int   6  7  5  1  6  4  2  4  4  1 ...
 $ LotSize     : num   612 449 631 501 473 687 515 583 483 607 ...
 $ HousePrice  : num  422180 328940 523640 379040 456860 ...
```

confirm for missing values

```
colSums(is.na(housing_data))
```

HouseSize	Bedrooms	Bathrooms	Age	Garage	DistancedCBD
0	0	0	0	0	0
CrimeRate	SchoolRating	LotSize	HousePrice		
0	0	0	0		

split data into training and testing

```
split_index <- createDataPartition(housing_data$HousePrice, p = 0.8, list = FALSE)
train <- housing_data[split_index, ]
test <- housing_data[-split_index, ]
```

create a training matrix

```
train_x <- as.matrix(train[, 1:9])
test_x <- as.matrix(test[, 1:9])
train_y <- train$HousePrice
test_y <- test$HousePrice
```

convert to Dmatrix

```
dtrain <- xgb.DMatrix(data = train_x, label = train_y)
dtest <- xgb.DMatrix(data = test_x, label = test_y)
```

train xgboost model

```
params <- list(objective = "reg:squarederror", eval_metric = "rmse", eta = 0.1, max.depth = 5,
               subsample = 0.8, colsample_bytree = 0.8)
xgb_model <- xgb.train(params = params, data = dtrain, nrounds = 150,
                      watchlist = list(train = dtrain),
                      verbose = 1)
```

[1] train-rmse:72690.413816
[2] train-rmse:66510.484761
[3] train-rmse:61013.162819
[4] train-rmse:55991.655681
[5] train-rmse:51596.068253
[6] train-rmse:49763.971223
[7] train-rmse:47952.735875
[8] train-rmse:44137.296399
[9] train-rmse:40602.703859
[10] train-rmse:37476.096060
[11] train-rmse:34629.566788
[12] train-rmse:32010.417961
[13] train-rmse:29705.899867
[14] train-rmse:27505.191024
[15] train-rmse:25535.001881
[16] train-rmse:23715.165577
[17] train-rmse:22769.250984
[18] train-rmse:21116.228389
[19] train-rmse:19660.797921
[20] train-rmse:18391.096239
[21] train-rmse:17142.507874
[22] train-rmse:16092.094823
[23] train-rmse:15067.661451
[24] train-rmse:14134.593329
[25] train-rmse:13623.055499
[26] train-rmse:12776.813677
[27] train-rmse:12000.837632
[28] train-rmse:11360.793591
[29] train-rmse:10911.585152
[30] train-rmse:10416.035777
[31] train-rmse:9845.649483
[32] train-rmse:9495.824894
[33] train-rmse:8969.438055
[34] train-rmse:8521.160097
[35] train-rmse:8094.317557
[36] train-rmse:7730.115250
[37] train-rmse:7370.276707
[38] train-rmse:7074.740495
[39] train-rmse:6773.458511
[40] train-rmse:6455.324636
[41] train-rmse:6179.074479
[42] train-rmse:5929.441172
[43] train-rmse:5700.269723
[44] train-rmse:5473.315415
[45] train-rmse:5262.526401
[46] train-rmse:5114.902348
[47] train-rmse:4925.636116
[48] train-rmse:4748.672345
[49] train-rmse:4589.524081

[50]	train-rmse:4450.358158
[51]	train-rmse:4327.107094
[52]	train-rmse:4209.299129
[53]	train-rmse:4093.905294
[54]	train-rmse:3993.324764
[55]	train-rmse:3879.647255
[56]	train-rmse:3788.660709
[57]	train-rmse:3686.502022
[58]	train-rmse:3577.045897
[59]	train-rmse:3493.344009
[60]	train-rmse:3420.366784
[61]	train-rmse:3349.806104
[62]	train-rmse:3277.526492
[63]	train-rmse:3217.678342
[64]	train-rmse:3137.086257
[65]	train-rmse:3084.552669
[66]	train-rmse:3005.255061
[67]	train-rmse:2953.546587
[68]	train-rmse:2891.990635
[69]	train-rmse:2842.458374
[70]	train-rmse:2791.669307
[71]	train-rmse:2749.213382
[72]	train-rmse:2692.935094
[73]	train-rmse:2660.258167
[74]	train-rmse:2612.874840
[75]	train-rmse:2574.304208
[76]	train-rmse:2521.889729
[77]	train-rmse:2479.857502
[78]	train-rmse:2444.916931
[79]	train-rmse:2414.879520
[80]	train-rmse:2379.106708
[81]	train-rmse:2359.087620
[82]	train-rmse:2312.788793
[83]	train-rmse:2279.633479
[84]	train-rmse:2251.273711
[85]	train-rmse:2221.666508
[86]	train-rmse:2189.731693
[87]	train-rmse:2169.925425
[88]	train-rmse:2139.927939
[89]	train-rmse:2108.373587
[90]	train-rmse:2095.158723
[91]	train-rmse:2063.823448
[92]	train-rmse:2040.866524
[93]	train-rmse:2020.521008
[94]	train-rmse:1989.274680
[95]	train-rmse:1958.747159
[96]	train-rmse:1936.394204
[97]	train-rmse:1913.313404
[98]	train-rmse:1897.851610
[99]	train-rmse:1876.138259

[100] train-rmse:1847.054681
[101] train-rmse:1835.765819
[102] train-rmse:1825.116004
[103] train-rmse:1801.992391
[104] train-rmse:1787.670165
[105] train-rmse:1762.171082
[106] train-rmse:1741.072680
[107] train-rmse:1722.303178
[108] train-rmse:1702.924013
[109] train-rmse:1683.658437
[110] train-rmse:1669.918049
[111] train-rmse:1645.809033
[112] train-rmse:1621.028913
[113] train-rmse:1607.471427
[114] train-rmse:1588.071869
[115] train-rmse:1562.658678
[116] train-rmse:1549.041993
[117] train-rmse:1524.454019
[118] train-rmse:1505.760682
[119] train-rmse:1490.060489
[120] train-rmse:1460.736322
[121] train-rmse:1442.511914
[122] train-rmse:1435.696765
[123] train-rmse:1411.417611
[124] train-rmse:1392.656872
[125] train-rmse:1380.509427
[126] train-rmse:1358.771020
[127] train-rmse:1342.910442
[128] train-rmse:1321.580278
[129] train-rmse:1311.200760
[130] train-rmse:1304.156046
[131] train-rmse:1278.731583
[132] train-rmse:1263.413469
[133] train-rmse:1241.761996
[134] train-rmse:1215.478675
[135] train-rmse:1189.755099
[136] train-rmse:1180.307077
[137] train-rmse:1160.587624
[138] train-rmse:1144.163710
[139] train-rmse:1133.878535
[140] train-rmse:1126.607024
[141] train-rmse:1113.979308
[142] train-rmse:1098.399382
[143] train-rmse:1085.317357
[144] train-rmse:1066.119455
[145] train-rmse:1058.915603
[146] train-rmse:1040.543873
[147] train-rmse:1023.712963
[148] train-rmse:1007.829376

```
[149] train-rmse:994.650150  
[150] train-rmse:983.593077
```

predict house price

```
predic <- predict(xgb_model, dtest)  
head(predic)
```

```
[1] 414530.7 237276.4 342235.4 421195.0 403231.3 411294.5
```

model evaluation

```
rmse_value <- rmse(test_y, predic)  
mae_value <- mae(test_y, predic)  
mse_value <- mse(test_y, predic)  
r2 <- cor(test_y, predic)^2  
cat("rmse:", rmse_value, "\n")
```

```
rmse: 12310.02
```

```
cat("mae", mae_value, "\n")
```

```
mae 9864.861
```

```
mse_value
```

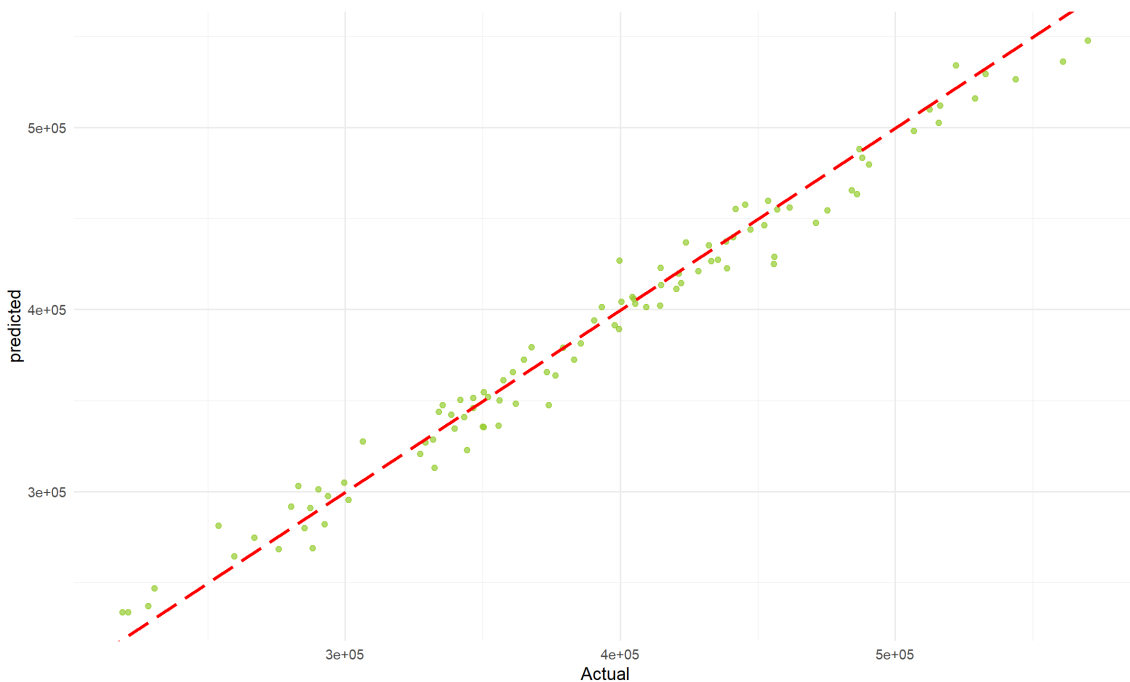
```
[1] 151536612
```

```
r2
```

```
[1] 0.9800824
```

scatter plot of actual vs predicted

```
ggplot(data.frame(Actual = test_y, predicted = predic), aes(Actual, predicted)) +
  geom_point(color = "yellowgreen", alpha = 0.7) + geom_abline(slope = 1, intercept = 0,
    colour = "red", linewidth = 1, linetype = "longdash") + theme_minimal()
```



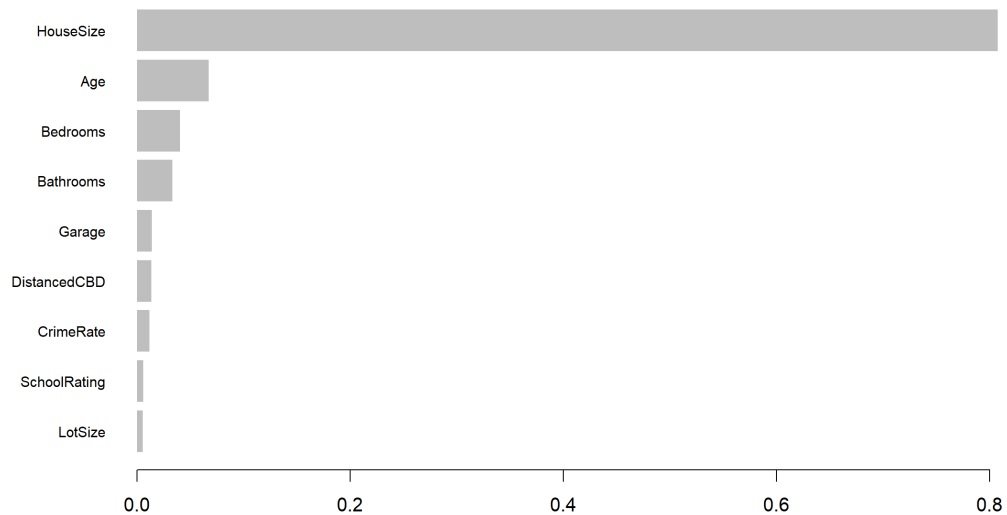
feature importance

```
importance <- xgb.importance(feature_names = colnames(train_x), model = xgb_model)
importance
```

	Feature <char>	Gain <num>	Cover <num>	Frequency <num>
1:	HouseSize	0.807301647	0.33183437	0.22919180
2:	Age	0.067265290	0.13045975	0.14414958
3:	Bedrooms	0.040628698	0.08136289	0.10012063
4:	Bathrooms	0.033503844	0.07835380	0.08443908
5:	Garage	0.014288845	0.06218096	0.06664656
6:	DistancedCBD	0.013643331	0.09370102	0.12484922
7:	CrimeRate	0.011941135	0.08443066	0.10072376
8:	SchoolRating	0.005932499	0.03617619	0.05247286
9:	LotSize	0.005494711	0.10150036	0.09740651

feature importance plot

```
xgb.plot.importance(importance_matrix = importance)
```



predict new instance

```
new_house <- matrix(c(2200, 4, 3, 8, 2, 10, 20, 8, 700), nrow = 1)
new_house <- xgb.DMatrix(new_house)
cat("predicted house model:", predict(xgb_model, new_house))
```

```
predicted house model: 493200.6
```